

COMP 10066

Week 9 Lecture

Bug Reporting and Resolution

Lecture Overview

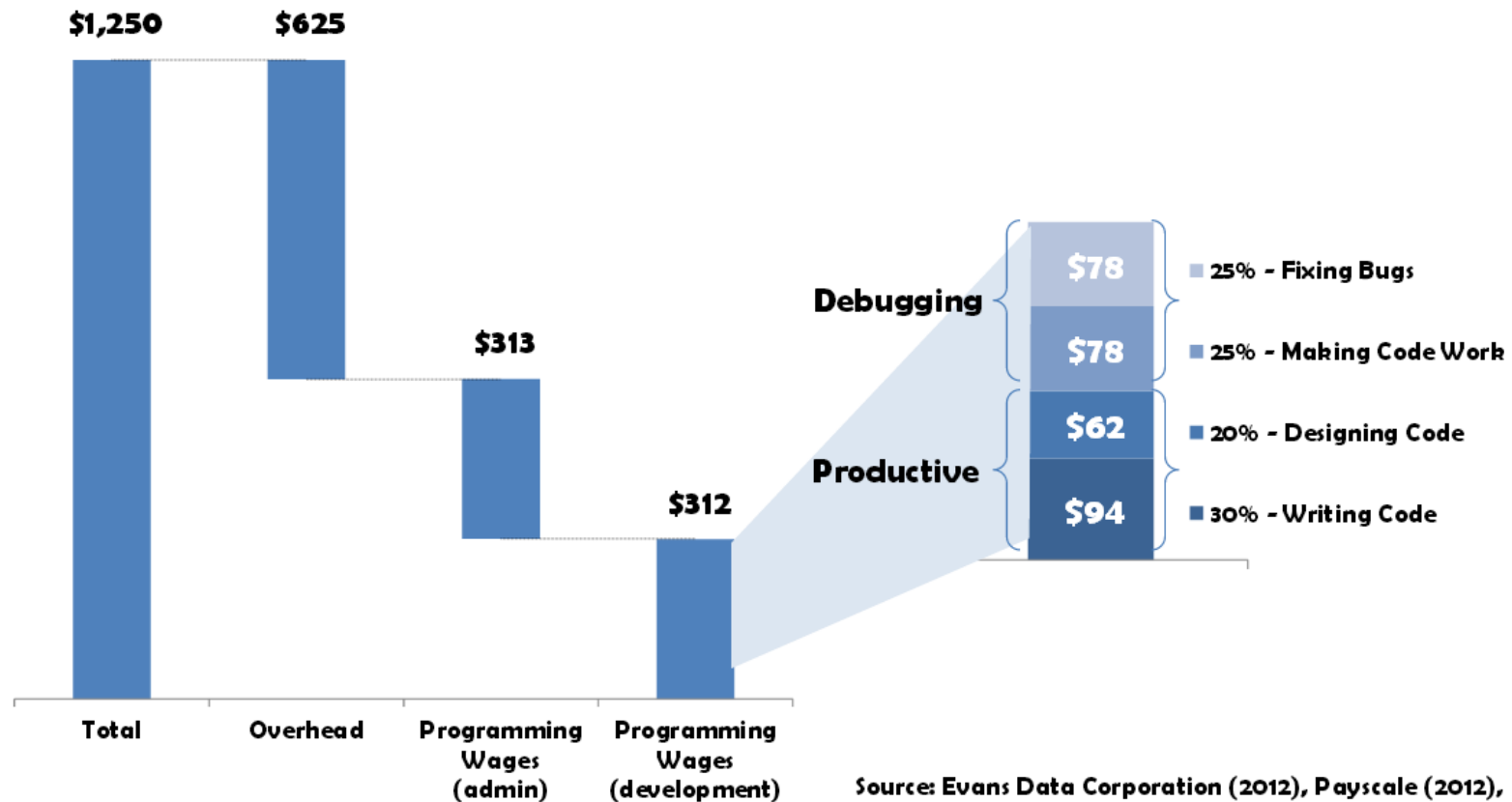
- Cost of Software Defects
- Reporting Bugs
- Bug Reporting Tracking
- Prioritizing Bugs

Cost of Bugs

- A interesting study:
 - "Cambridge University Study States Software Bugs Cost Economy \$312 Billion Per Year"
 - <http://www.prweb.com/releases/2013/1/prweb10298185.htm>
 - **Programmers spend 50% of time debugging**

RESULTS: The global cost of software development is US\$1.25 trillion

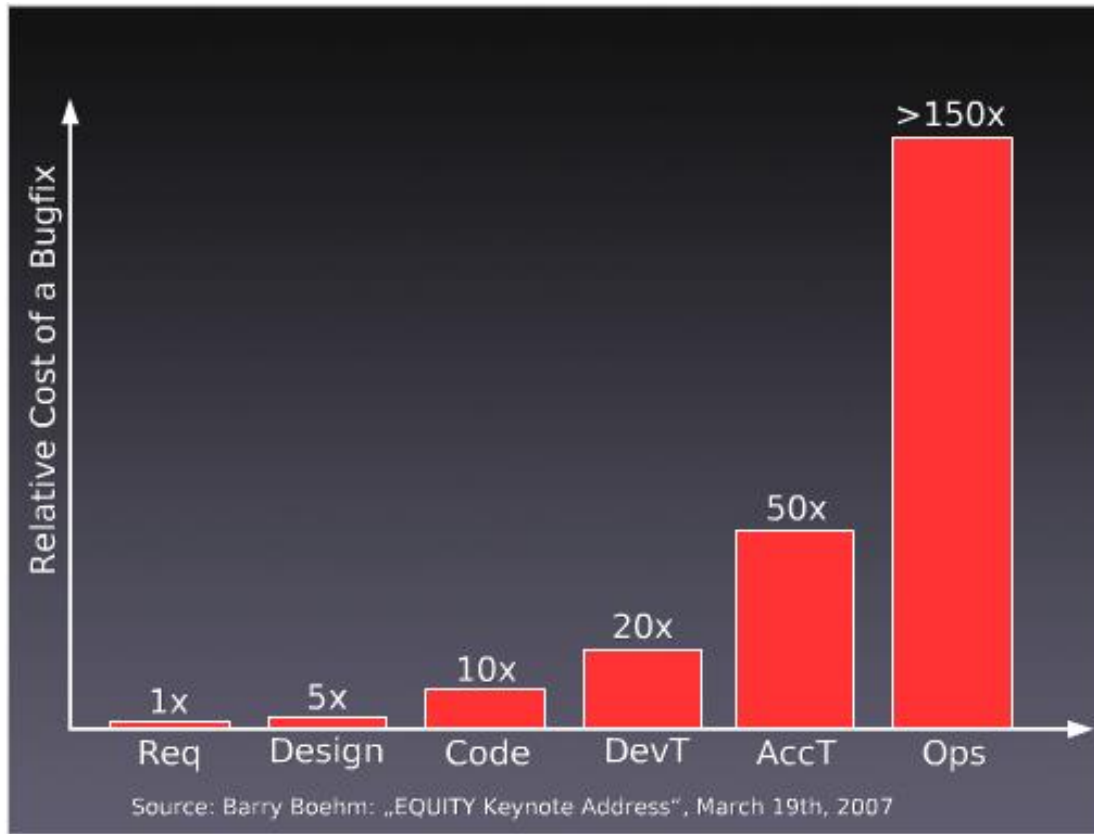
Software development cost structure (US\$ billion)



reference:

<http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.370.9611&rep=rep1&type=pdf>

Relative Bug Cost



Legend:

Req –
Requirements

DevT –
Development
Testing

AccT –
Acceptance
Testing

Ops –
In Operation

** some studies suggest
these numbers are a little
high

Cost Of Bugs

- Are the costs really that high?
- Some argue the cost per defect calculation is not accurate
- See Cost Per Defect PDF on MyCanvas

Bug Reporting and Resolution

- Two important aspects of bug reporting that will increase the chances of the bug being corrected:
 - When it is reported
 - An effective description
 - Provide a procedure to duplicate the bug
 - Video is sometimes used to record steps
 - <http://www.usertesting.com/>

Attributes for an Effective Bug Description

- Provide information that is relevant to the bug.
 - OS Version, **Software Build**
- One bug / report
 - Extra information can cause confusion.
 - Do not report more than one bug at a time.
 - Can reference another bug if there appears to be a relationship
- Document the process to reproduce the bug
 - Create as simple a case as possible to show bug.
 - Step-by-Step
 - Confirm the procedure before submitting bug.

Bug Report Content from the Tester Perspective

- Program Identification
- Version Information
- Date
- Reported by
- Suspected type of bug (Coding, Design, User Interface, Documentation)
- Problem Summary
 - Reproducible (Y/N)
 - Steps to reproduce
 - Suggestions for correction

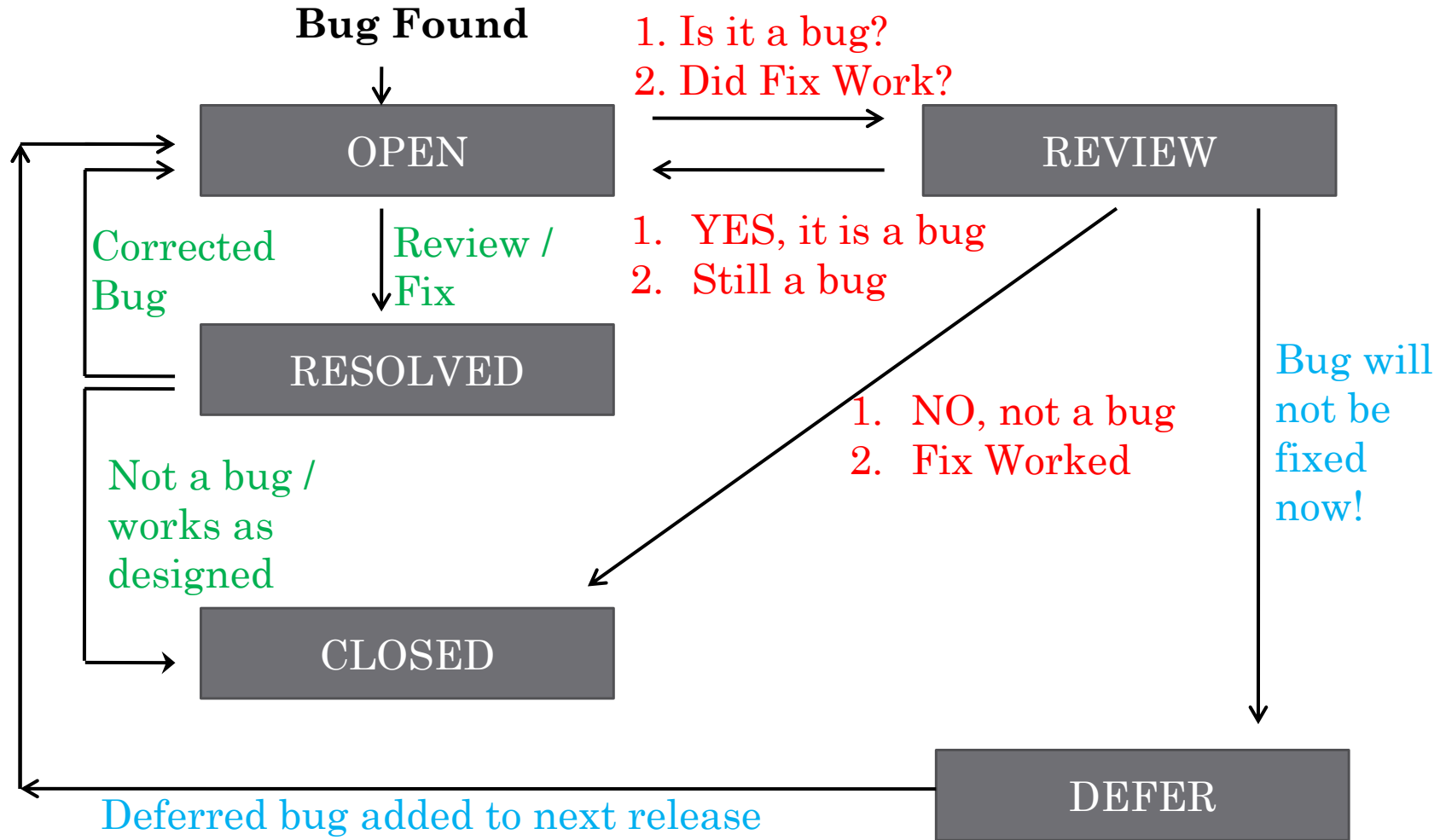
Bug Report Content from the Developer Perspective

- Assigned to
- Status (Open/Closed)
- Priority
- Resolution
 - Pending, Deferred, Withdrawn, Fixed, As Designed, **Need more Info, Can't reproduce, Can't be fixed!**
- Resolved by with Date
- Solution Tested and confirmed with Date
 - Version / Build should be included with the fix

Examples of Bug Summaries

- Bug Summary Contents vary by company
- <https://www.softwaretestinghelp.com/sample-bug-report/>
- <https://usersnap.com/blog/bug-report-template/>

Example Bug Workflow



Green – Programmer

Blue – Project Management

Red - Tester

Responsibilities

- The programmer is responsible to confirm the bug report:
 - **Not a Bug (as designed) – Move to Close**
 - Bug – Defer or Resolve

The tester must confirm the bug resolution or fix:

- **The applied fix worked - Move to Close**
- Still a Bug – Fix did not work/incomplete - Move back to Open

Reasons bug fixes are delayed or not fixed

- Why do you think bugs are delayed or not fixed?

Reasons bug fixes are delayed or not fixed

1. Not enough time
2. Not really a bug
3. Too Risky to Fix
4. Not worth it to fix – impacts few users
5. Bug report is vague – difficult to reproduce

Bug Priorities

- Based on the information given about the bug and how severe the bug is, the bug is often assigned a priority
- Low priority bugs are delayed or not fixed
- High priority bugs are typically given immediate attention
- For this week's lecture handout, review the 6 bugs given and decide on the order you would rank them in terms of priority
 - Can any of these bugs be delayed?

Lecture Handout

Practicing Writing a Bug Report

- Bug Summary should include:
 - Bug Identifier / Number
 - Test Case Identifier / Number the bug was found in
 - Bug Description
 - Steps to reproduce the bug, including the exact data inputted into the program
 - The actual result obtained
 - The expected result – what you expected to happen
 - Opinion of what went wrong if you can assume the issue
 - Screenshot of the issue if possible

Practicing Writing a Bug Report

- Let's take the Black Box Testing Example we did last lecture
- Can you remember any Test to Pass bugs in this piece of software?

COMP10066 Black Box Lecture Example - Strings

Enter a String
(Can include spaces and any keyboard characters)

Summarize Clear Exit

of Words =

of upper case Letters =

of lower case Letters =

of spaces between words =

Most frequent letter(count) =

Practicing Writing a Bug Report

- Let's walk through writing a good bug description for one of these bugs
 - Steps to reproduce the bug, including the exact data inputted into the program
 - The actual result obtained
 - The expected result – what you expected to happen
 - Opinion of what went wrong if you can assume the issue
 - Screenshot of the issue if possible
- Use the example created in class as an example of what is expected on future assignments